Dart Programming – Loop Assignments

Beginner-friendly assignments using for, while, do-while, and nested loops.

1. Multiplication Table ■

Ask the user for a number and print its multiplication table from 1 to 10.

Expected pattern/output:

```
Example for 7:
7 x 1 = 7
7 x 2 = 14
...
7 x 10 = 70
```

Concepts: for loop, variables, arithmetic

2. Reverse Number Pattern ■

Print the following pattern for n = 5:

Expected pattern/output:

```
12345
1234
123
12
1
```

Concepts: nested loops, decrementing loops

3. Number Pyramid ■■

Take n as input and print a growing number pattern.

Expected pattern/output:

```
1
12
123
1234
12345
```

Concepts: nested for loops, pattern generation

4. Number Triangle ■■

Print a pattern where each row contains the row number repeatedly.

Expected pattern/output:

```
1
22
333
4444
```

```
55555
```

Bonus: Modify it to print 1, 12, 123, 1234, 12345.

5. Star Pyramid ■■

Print a centered pyramid of stars for n = 5.

Expected pattern/output:

```
*
***
*****
*******
*********
```

Challenge: Use nested loops to generate both spaces and stars.

6. Prime Numbers ■■■

Write a program that prints all prime numbers between 1 and 100.

Expected pattern/output:

```
2
3
5
7
11
13
17
19
...
```

Bonus: Let the user enter the starting and ending numbers. Concepts: nested loops, if-else, % operator.